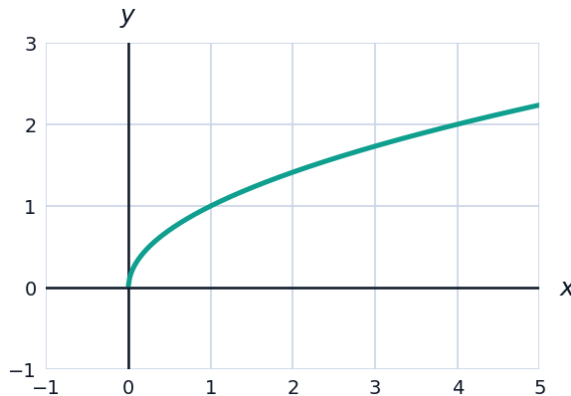


Area Between Curves



Area between two curves

- 1 Find the area of the region bounded by $y = x^2$ and $y = x + 2$.
- 2 The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.



Setting up with respect to y, and total area

- 3 Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y .
- 4 Find the total area between $f(x) = \sin x$ and the x -axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).

More area problems

- 5 Find the area of the region bounded by $y = x^2 - 4$ and $y = 4 - x^2$.
- 6 Find the area between $y = x^3$ and $y = x$ on $[0, 1]$ (note that $y = x$ is above $y = x^3$ on this interval).

Areas requiring case splits

- 7 Find the area between $y = x^3$ and $y = x$ on $[-1, 1]$ (note the curves cross at $x = 0$, so the identity of the upper curve switches).
 - 8 Find the area enclosed between $y = 2x$ and $y = x^2$ on the interval where $y = 2x$ is above $y = x^2$.
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Areas involving trigonometric and exponential curves

- 9 Find the area between $y = \cos x$ and $y = 0$ on $[0, \frac{\pi}{2}]$.
- 10 Find the area between $y = e^x$ and $y = e^{-x}$ on $[0, 1]$ (note $e^x \geq e^{-x}$ on this interval).